732A98 Visualization
Division of Statistics and Machine Learning
Department of Computer and Information Science

Laboratory work 3

Instructions

- The lab report should be submitted via LISAM before the deadline. The submission should be an RMD file, HTML file and all supporting files compressed into one ZIP archive.
- Create a report for the lab solutions in RMarkdown. Make sure that it can be compiled to HTML and that all paths in RMD file are relative to the current directory where the RMD file is located. **Reports that can not be compiled are returned without revision.**
- **Include all your codes as an appendix into your report; Also show the corresponding parts of the code in the flowing text of the report.**
- A typical lab report should contain 2-4 pages of text plus some figures and the appendix with the code.
- Be concise and do not include unnecessary printouts and figures produced by the software that are not required in the assignments.

Assignment 1

The file **prices-and-earnings.txt** shows UBS's (one of the largest banks in the world) report comparing prices, wages, and other economic conditions in cities around the world. Some of the variables measured in 73 cities are Cost of Living, Food Costs, Average Hourly Wage, average number of Working Hours per Year, average number of Vacation Days, hours of work (at the average wage) needed to buy an iPhone, minutes of work needed to buy a Big Mac, and Women's Clothing Cost.

1. For the analysis, import the data to R and keep only the columns with the following numbers: 1,2,5,6,7,9,10,16,17,18,19. Use the first column as labels.
2. Plot a heatmap of the data without doing any reordering. Is it possible to see clusters or outliers?
3. Compute distance matrices by a) using Euclidian distance and b) using one minus correlation. For both cases, compute orders that optimize the Hamiltonian Path Length and use Hierarchical Clustering (HC) as the optimization algorithm. Plot the two respective heatmaps and state which plot seems to be easier to analyze and why. Make a detailed analysis of the plot based on Euclidian distance. **Use Euclidian Distance matrix in all coming steps**.
4. Compute a permutation that optimizes Hamiltonian Path Length but uses Traveling Salesman Problem (TSP) as solver. Compare the heatmap given by this reordering with the heatmap produced by the HC solver in the previous step – which one seems to be better? Compare also the objective function values such as Hamiltonian Path length and Gradient measure achieved by row permutations of TSP and HC solvers (Hint: use the criterion() function)
5. Use the data obtained by using the HC solver and create a radar chart diagram with juxtaposed radars. Identify two smaller clusters in your data (choose yourself which ones) and the most distinct outlier.
6. Which of the tools you have used in this assignment (heatmaps or radar charts) was best in analyzing these data? From which perspective? (e.g. efficiency, simplicity, etc.)

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Assignment 2

The files **trainData.dat** and **trainMeta.dat** contain information about a network of suspected individuals involved in the bombing of commuter trains in Madrid on March 11, 2004. The file **trainMeta.dat** contains the names of individuals (first column) and Bombing group (second column) which shows "1" if person participated in placing the explosives and "0" otherwise. According to the order in this file, individuals were enumerated 1-70. The file **trainData.dat** contains information about connections between the individuals (first two columns) and strength of ties linking (from one to four): 1 = Trust-friendship (contact, kinship, links in the telephone center), 2 = Ties to Al Qaeda and to Osama Bin Laden, 3 = Co-participation in training camps and/or wars, 4 = Co-participation in previous terrorist Attacks (Sept 11, Casablanca).

1. Use the visNetwork package to plot a graph in which
   a. you use strength on the links variable
   b. nodes are colored by Bombing Group
   c. size of nodes is proportional to the number of connections (the function strength() from IGRAPH might be useful here)
   d. you use a layout that optimizes repulsion forces (visPhysics(solver="repulsion")).
   e. all nodes that are connected to a currently selected node by a path of length of one are highlighted

   Analyze the obtained network, in particular describe which clusters you see in the network.
2. Add a functionality to the plot in step 1 that highlights all nodes that are connected to the selected node by a path of length one or two. Check some of the largest nodes and comment on which individual had the best opportunity to spread the information in the network. Read some information about this person on Google and present your findings.
3. Compute clusters by optimizing edge betweenness and visualize the resulting network. Comment on whether the clusters you identified manually in step 1 were also discovered by this clustering method.
4. Use adjacency matrix representation to perform a permutation by Hierarchical Clustering (HC) seriation and visualize the graph as a heatmap. Find the most pronounced cluster and comment whether this cluster was also discovered in steps 1 and 3.

Submission procedure

When submitting the report, remember to **specify in LISAM your group name** and all group members that wrote the report!

Make sure that you or your group mate submits the group report using *Lab X* item in the *Submissions* folder before the deadline. Make sure that **the report contains a Statement of Contribution** describing how each group member has contributed into the group report

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